

AcePharm — Mobile App: Project Tracker

Companion Cross-Platform App Tracker for iOS & Android (Flutter / Dart)

This document tracks the development stages and QA milestones for the **AcePharm** native mobile companion app. Check off each stage as the project progresses to ensure UI token fidelity, question engine accuracy, offline resilience, and store deployment readiness.

Phase 1: Setup, Flutter Architecture & Foundation

- **Project Scaffolding:** Initialize Flutter 3.x with clean layered architecture (core, data, domain, presentation) building on established Takween mobile patterns.
- **Design System Theme Tokens:** Translate AcePharm tokens into Flutter ThemeData: Ace Indigo (#4F46E5), Deep Indigo (#3730A3), Ink (#111827), Slate (#64748B), 8pt grid, and 44x44pt touch targets.
- **Dependency & Storage Configuration:** Configure Dio networking, flutter_secure_storage for auth tokens, Hive / SQLite for offline attempt storage, and Firebase Core.
- **Platform Identifiers & Permissions:** Configure unique bundle IDs and permissions for iOS (Runner.xcworkspace) and Android (build.gradle.kts).
 - **Test Environment:** iOS Simulator & Android Emulator (verify clean compile, design token rendering, and platform bundle IDs).

Phase 2: Content & Unified API Data Layer

- **API Contract Alignment:** Map mobile data contracts against the web platform's Next.js API endpoints for sessions, questions, categories, and progress.
- **Domain Modeling:** Implement strongly typed Dart models: Question, Option, Explanation, SubtopicNote, AttemptPayload, UserProfile, and TopicProgress.
- **Mock Fixture Engine:** Deploy structured JSON mock fixtures matching Seed Question Bank v0.1 (135 questions) enabling rapid mobile development prior to web Milestone 3 deployment.
- **Unit & Model Serialization Tests:** Verify JSON serialization/deserialization, calculation formula parsing, and model immutability.
 - **Test Environment:** Headless Unit Test Runner (dart test / flutter test) verifying serialization and mock data contracts.

Phase 3: Authentication & Offline State Layer

- **Native Authentication UI:** Build login, registration, and password reset screens integrating with web Auth.js token endpoints.
- **Secure Token Lifecycle:** Store JWT access/refresh tokens in iOS Keychain and Android Keystore with automatic token refresh interceptors.
- **5-Step Onboarding Flow:** Touch-optimized onboarding: Stage (MPharm Y2-4 / Trainee), Exam Date, Goal, Daily Target, and School of Pharmacy.
- **Reactive State Architecture:** Implement state controllers (Riverpod/Bloc) managing auth, session configurations, and progress state.
 - **Test Environment:** Physical iOS & Android Devices (verify real keychain token persistence, bio-auth, and session resume).

Phase 4: Session Builder & Question Practice (Core Feature)

- **Mobile Session Builder:** Intuitive bottom sheets and category selectors for therapeutic topics, high-risk drugs, question counts, and modes (Learn vs. Timed).
- **Question Screen & Strict Answering Rules:** Confidence rating prompt (Low, Medium, High) captured before submission. Zero color feedback prior to tapping Submit. Distinct icon, label, and color feedback after submission. No auto-advance.
- **Clinical Explanations & Notes:** Collapsible option rationales for all 5 choices, comprehensive subtopic clinical notes, personal note editor, bookmarks, and question reporting attributed to 'AcePharm Clinical Editorial Team'.
- **Widget Testing & Formula Rendering:** Automated widget tests for question layout, Geist Mono math formula formatting, and touch target validation.
 - **Test Environment:** Simulators & Physical Devices (verify rapid one-handed answering, formula display, and question navigation).

Phase 5: Contextual AI Tutor ('Ask Ace') & Progress

- **Ask Ace Bottom Sheet:** Contextual AI entry point docked beneath explanations, enabling learners to ask clarifying clinical or calculation questions.
- **Streaming Grounded Responses:** Stream responses generated exclusively from AcePharm's reviewed content with explicit source citations. Refuse out-of-domain queries cleanly.

- **Five-Metric Dashboard View:** Render First-Attempt Accuracy (prominent), Practice Accuracy, Repeat Accuracy, Confidence Calibration, and Curriculum Coverage.
- **Mobile Topic Map:** Interactive visual map using the 6-state progression: Not Started → First Pass → Needs Attention → Developing → Secure → Due for Review.
 - **Test Environment: Physical Devices** (verify streaming AI tokens, markdown/math rendering in chat sheet, and topic map touch targets).

Phase 6: Offline Resilience & Hospital Ward Reliability

- **Offline Attempt Queue:** When connectivity drops, queue submitted question attempts locally in SQLite/Hive with accurate timestamps and retry automatically when online.
- **Cached Study Assets:** Cache active question sessions, clinical notes, and dashboard summaries locally so revision continues uninterrupted without internet.
- **Crash-Proof Network Handling:** Ensure zero crashes, blank screens, or lost answers on network timeouts; display non-intrusive offline status banner.
- **Responsive Matrix Verification:** Validate layouts across compact phones (iPhone SE, small Androids) and large tablets (iPad, Galaxy Tab).
 - **Test Environment: Physical iOS & Android Devices in Airplane Mode** (simulate complete offline session answering and background sync restoration).

Phase 7: Store Packaging, QA & Submission

- **Full Regression QA Pass:** Verify complete revision journey: login, 20-question session, confidence capture, explanation reading, Ask Ace query, progress update, and offline sync.
- **Subscription Status Display:** Display web-purchased subscription status (Plan name, renewal date, price) adhering to Apple App Store Reader exceptions.
- **Store Release Packaging:** Generate signed Android App Bundle (AAB) for Google Play Console Internal Testing and iOS Archive for Apple TestFlight.
- **Submission Deliverables:** Record 2–4 minute Loom walkthrough video and complete the 12-point Mobile Submission Document with the Path to Production plan.
 - **Test Environment: Google Play Console Internal Testing Track & Apple TestFlight Sandbox** (clean installation on unprovisioned physical devices).

Project Links & Distribution Status

- **GitHub Repository:** <https://github.com/takweentutors1/acepharm-web-app> (mobile-app directory)
- **Google Play Console (Internal Track):** Android AAB upload pending website verification clearance
- **Apple TestFlight:** Pending Apple Developer Program account setup
- **Loom Walkthrough Video:** Record 2–4 min mobile demonstration upon milestone completion
- **Release Binaries:** v0.1.2-release.apk ready in mobile-app/release/